In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.ticker import AutoMinorLocator
import sklearn as sk
from scipy import stats as st
from scipy.optimize import minimize
import pandas as pd
```

In [2]:
```
np.set_printoptions(formatter={'float': lambda x: "{0:0.3e}".format(x)}) #tc
```

Functions

In [3]:
```
#for plotting distributions
def plot_distrib(data, bins = 100, title = ""):
    plt.hist(data, bins=bins, density=True)
    plt.title(title)
    plt.ylabel("Probability density")
```

In [4]:
```
#to calculate 1σ uncertainty bands and medians
def median_and_error(data):
    med, low, high = np.percentile(data, [50, 16, 84])
    median_and_error = np.array([med, low - med, high - med])

    return median_and_error
```

In [5]:
```
#Fits a skew-normal distribution to a given set of percentiles.
def fit_skewnorm(percentiles, values): #the percentiles and corresponding va

    #set initial guesses as if it was a gaussian
    median = np.interp(0.5, percentiles, values)
    p10 = np.interp(0.1, percentiles, values)
    p90 = np.interp(0.9, percentiles, values)
    init_scale = (p90 - p10) / 2.56 if (p90 - p10) > 0 else 1  #for a gaussi


    initial_guesses = [0., median, init_scale]

    bounds = [(-20, 20), (None, None), (1e-6, None)] #(a, loc, scale) -> kee

    #Error function to minimize
    def error_function(params):
        a, loc, scale = params
        theoretical_values = st.skewnorm.ppf(percentiles, a=a, loc=loc, scal
        return np.sum((theoretical_values - values) ** 2) #sum of squared er

    result = minimize(
        error_function,
        initial_guesses,
        method='SLSQP',
        bounds=bounds,
        options={'ftol': 1e-9, 'disp': False}
    )
```

```
    return result.x #(a, loc, scale)
```

Isolation Time Figure

This notebook calculates and plots isolation time distributions for 92Nb and 146Sm using abundance ratios from the CCSN models from Ritter et al. (2018) and Roberti et al. (2024) and the SNIa ratios from Lugaro et al. (2016). This work is largely based on re-implementing the *Mathematica* notebook 'isolation_time_figures.nb' from Leckenby et al. (2024) in *Python* and expanding its functionality to account for mean-life uncertainties and modeling the GCE stochastic uncertainties described in Côte et al. (2019) by fitting a skew-normal distribution to the medians and confidence intervals they provide.

In [6]:
```
k = 2.3 #best
```

In [7]:
```
#production ratio
prIaNb = 1.58*10**-3
prIaSm =  0.347

EssNb = (3.2 * 10**-5, 0.3 * 10**-5)
EssSm = (8.28 * 10**-3, 0.44 * 10**-3)

#mean-life
tauNb = np.array([34.7, 2.4]) / np.log(2.) #from NUBASE2020
tauSm = np.array([71.74, 7.39]) / np.log(2.) #from Tang et al. (2025)
tau = np.array([tauNb,tauSm])

#age of the galaxy
TGal = 8400
```

In [8]:
```
#τ for table
print(tau)
```

```
[[5.006e+01 3.462e+00]
 [1.035e+02 1.066e+01]]
```

Fit skewnormal distribution to percentiles

Assuming that a skew-normal distribution is a sufficient approximation for the assymetric distributions of the ISS ratios.

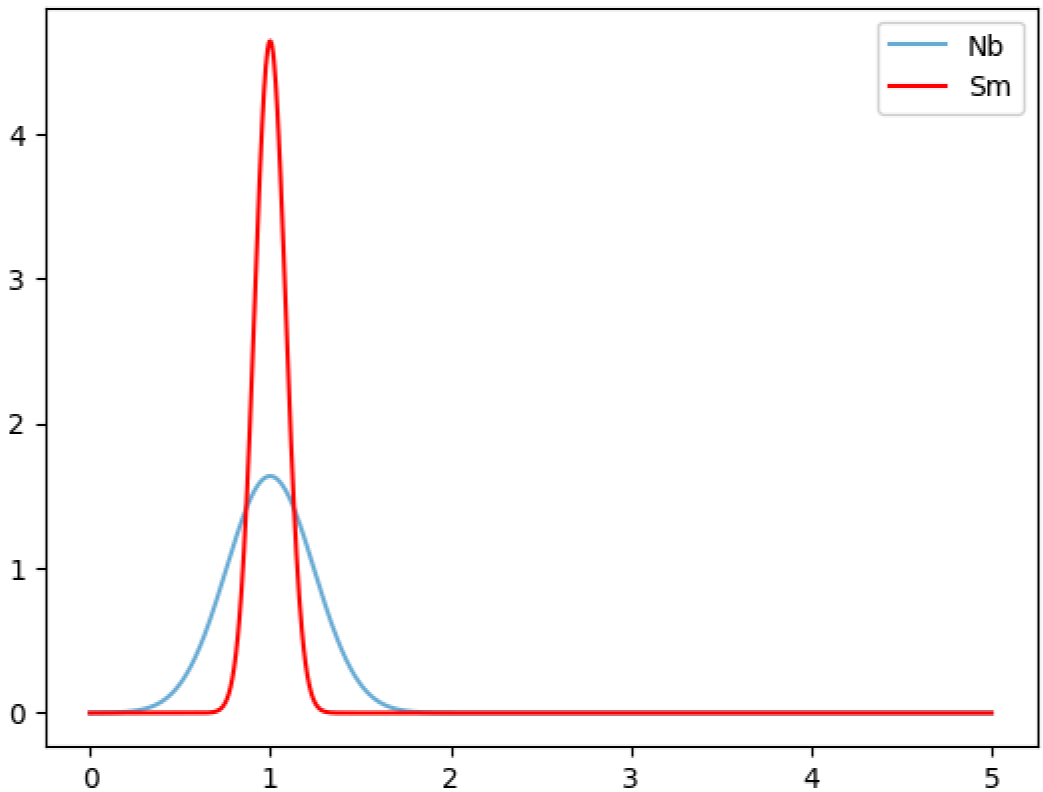
Percentiles from Table 3 in Côté et al. (2019), γ is taken to be 10 Myr.

In [9]:
```
percentiles = [0.16, 0.50, 0.84]

statNb = [3.13-0.73, 3.13, 3.13+0.80]
statSm = [9.99-0.84, 9.99, 9.99+0.87]

paramNb = fit_skewnorm(percentiles, statNb)
```

```
paramSm = fit_skewnorm(percentiles, statSm)

distNb = st.skewnorm(paramNb[0], paramNb[1], paramNb[2])
distSm = st.skewnorm(paramSm[0], paramSm[1], paramSm[2])

#normalise
paramNb[1:] = paramNb[1:] / distNb.mean()
paramSm[1:] = paramSm[1:] / distSm.mean()

#create normalised distribution
distNb = st.skewnorm(paramNb[0], paramNb[1], paramNb[2])
distSm = st.skewnorm(paramSm[0], paramSm[1], paramSm[2])
```

In [10]:
```
#visualising the fitted distributions
x = np.linspace(0,5,1000)
plt.plot(x,distNb.pdf(x=x),marker='none',label='Nb',color=plt.get_cmap('Blue
plt.plot(x,distSm.pdf(x=x),marker='none',label='Sm',color='r')
plt.legend()
```

Out[10]: <matplotlib.legend.Legend at 0x77ce40a983e0>



Calculate and plot isolation times for TypeIa prodcuction

Then I draw Monte Carlo distributions of the ESS and ISM ratios, and mean-lives. τ and ESS is assumed to be Gaussian, ISM comes from the fitted skew-normal distributions.

In [11]:
```
mcNum = 10**7 #number of samples for MC
```

In [12]:
```
#Sampling distributions for the τ values
ratTauNb = np.clip(np.random.normal(loc = tauNb[0], scale = tauNb[1], size=m
ratTauSm = np.clip(np.random.normal(loc = tauSm[0], scale = tauSm[1], size=m
```

In [13]:
```
#The ISM ratios
IsmNb = k * prIaNb * ratTauNb / TGal
IsmSm = k * prIaSm * ratTauSm / TGal
```

In [14]:
```
#Sampling distributions for the ISM values
ratIsmNb = np.clip(IsmNb * distNb.rvs(mcNum), 1e-30, np.inf)
ratIsmSm = np.clip(IsmSm * distSm.rvs(mcNum), 1e-30, np.inf)
```

In [15]:
```
#Sampling normal distribution for the ESS values
ratEssNb = np.clip(np.random.normal(loc = EssNb[0], scale = EssNb[1], size=m
ratEssSm = np.clip(np.random.normal(loc = EssSm[0], scale = EssSm[1], size=m
```

In [16]:
```
#Calculating isolation time
isoTNb = -np.log(ratEssNb / ratIsmNb) * ratTauNb
isoTSm = -np.log(ratEssSm / ratIsmSm) * ratTauSm
```

In [17]:
```
#downsample for visualisation (this is for optimisation)
isoTSampleNb = sk.utils.resample(isoTNb, n_samples=1000000)
isoTSampleSm = sk.utils.resample(isoTSm, n_samples=1000000)
```

In [18]:
```
#create the approximate isolation time distributions
isoTSampleNbKern = sk.neighbors.KernelDensity(kernel='gaussian', bandwidth=1
isoTSampleNbKern.fit(isoTSampleNb.reshape(-1, 1))
isoTSampleSmKern = sk.neighbors.KernelDensity(kernel='gaussian', bandwidth=1
isoTSampleSmKern.fit(isoTSampleSm.reshape(-1, 1))
```

Out[18]:

▾ KernelDensity i ?

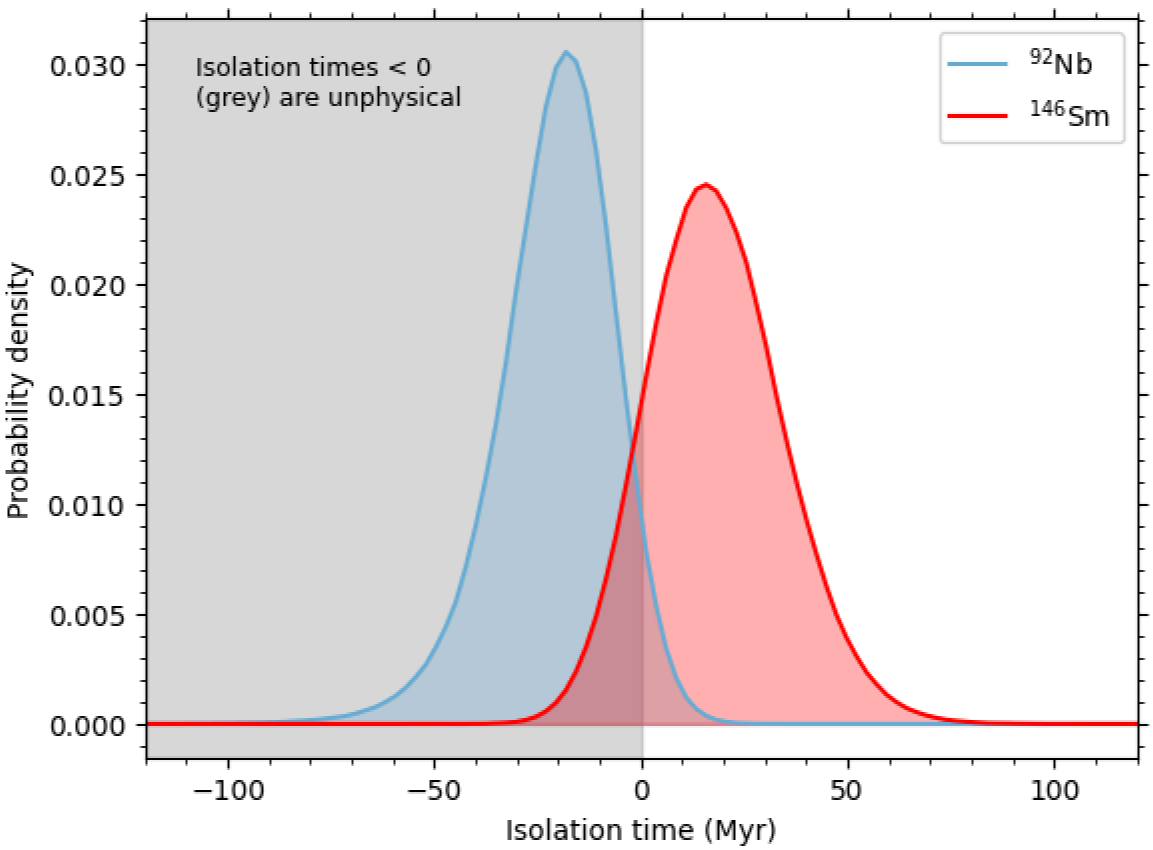
▶ Parameters

In [19]:
```
x = np.linspace(-120, 120, 100).reshape(-1, 1)
```

In [20]:
```
#evaluate the distribution at 100 points
isoTSampleNbDensity = np.exp(isoTSampleNbKern.score_samples(x))
isoTSampleSmDensity = np.exp(isoTSampleSmKern.score_samples(x))
```

In [21]:
```
#Creating figure
blue = plt.get_cmap('Blues')(2/4)

plt.plot(x, isoTSampleNbDensity, color=blue, label='$^{92}$Nb')
plt.fill_between(x.flatten(), isoTSampleNbDensity, 0, color=blue,  alpha=0.3

plt.plot(x, isoTSampleSmDensity, color='red', label='$^{146}$Sm')
plt.fill_between(x.flatten(), isoTSampleSmDensity, 0, color='red',  alpha=0.

plt.xlabel('Isolation time (Myr)')
plt.ylabel('Probability density')
```

```
plt.legend(loc='upper right')

#for tickmarks
plt.gca().xaxis.set_minor_locator(AutoMinorLocator())
plt.gca().yaxis.set_minor_locator(AutoMinorLocator())
plt.gca().tick_params(axis='x', which='both', top=True)
plt.gca().tick_params(axis='y', which='both', right=True)

plt.legend(loc='upper right')

#for the gray box
plt.axvspan(xmin=x.flatten()[0], xmax=0, color='gray', alpha=0.3, zorder=0)
plt.text(0.05, 0.95, "Isolation times < 0 \n(grey) are unphysical", transfor

plt.xlim(x[0], x[-1])

plt.savefig("../figures/"+"isoT_Ia.png",bbox_inches='tight')
plt.show()
```



Calculate and plot isolation times for CCSN prodcuction

In [22]:
```
# Import production ratios of different CCSN models
data = pd.read_csv("../data/production_ratios.txt", sep='\\s+')
data = data.set_index("model")
```

In [23]:
```
shock = np.array(data['initial_shock'])
mtot = np.array(data['MTOT'])
mrem = np.array(data['MREM'])
msurf = np.array(data['MSURF'])
modelSet = np.array(data['Set'])

prCCSnNb = np.array(data['Nb-92/Mo-92'])
prCCSnSm =  np.array(data['Sm-146/Sm-144'])
```

In [24]:
```
mcNum = 10**7 #number of samples for MC
```

In [25]:
```
#Sampling distributions for the τ values
ratTauNb = np.clip(np.random.normal(loc=tauNb[0], scale=tauNb[1], size=mcNum
ratTauSm = np.clip(np.random.normal(loc=tauSm[0], scale=tauSm[1], size=mcNum
```

In [26]:
```
#The ISM ratios
IsmNbCCSn = k * prCCSnNb[:, np.newaxis] * ratTauNb / TGal
IsmSmCCSn = k * prCCSnSm[:, np.newaxis] * ratTauSm / TGal
```

In [27]:
```
ratIsmNbCCSn = np.clip(IsmNbCCSn * distNb.rvs(mcNum), 1e-30, np.inf)
ratIsmSmCCSn = np.clip(IsmSmCCSn * distSm.rvs(mcNum), 1e-30, np.inf)

isoTNbCCSn = -np.log(ratEssNb/ ratIsmNbCCSn) * ratTauNb
isoTSmCCSn = -np.log(ratEssSm / ratIsmSmCCSn) * ratTauSm
```

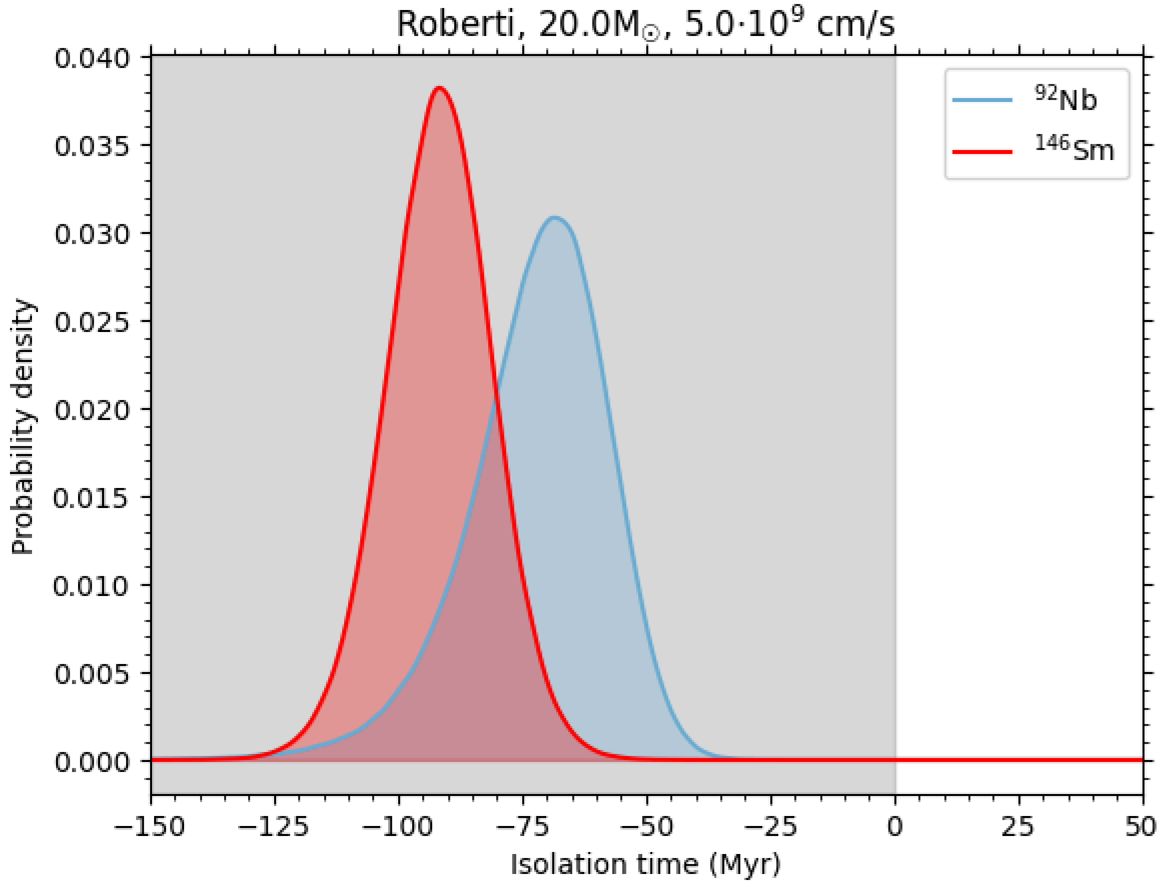
In [28]:
```
isoTSampleNbCCSn = []
isoTSampleSmCCSn = []

for i in range(np.size(prCCSnNb)):
    isoTSampleNbCCSn.append(sk.utils.resample(isoTNbCCSn[i], n_samples=10000
    isoTSampleSmCCSn.append(sk.utils.resample(isoTSmCCSn[i], n_samples=10000
```

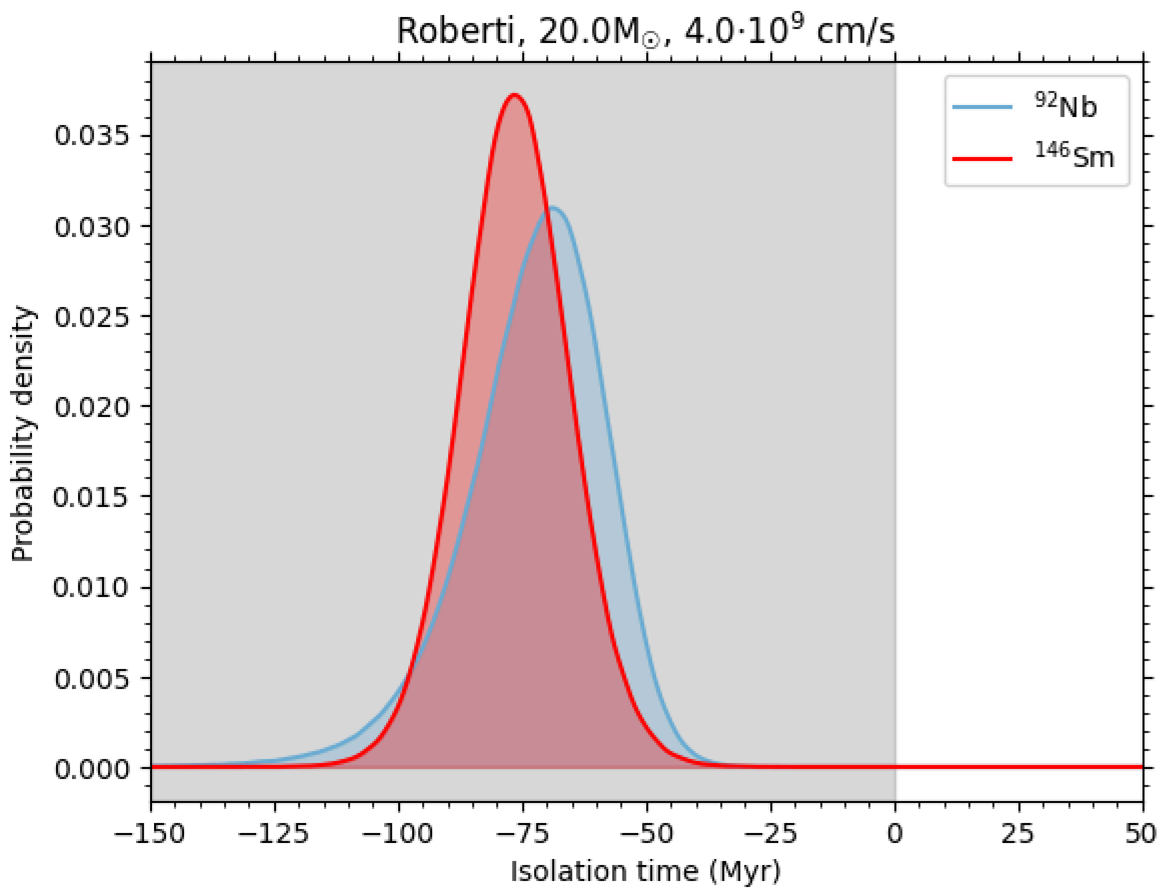
In [29]:
```
isoTNbKernCCSn = []
isoTSmKernCCSn = []

for i in range(np.size(prCCSnNb)):
    isoTNbKernCCSn.append(sk.neighbors.KernelDensity(kernel='gaussian', banc
    isoTNbKernCCSn[i].fit(isoTSampleNbCCSn[i].reshape(-1, 1))
    isoTSmKernCCSn.append(sk.neighbors.KernelDensity(kernel='gaussian', banc
    isoTSmKernCCSn[i].fit(isoTSampleSmCCSn[i].reshape(-1, 1))
```

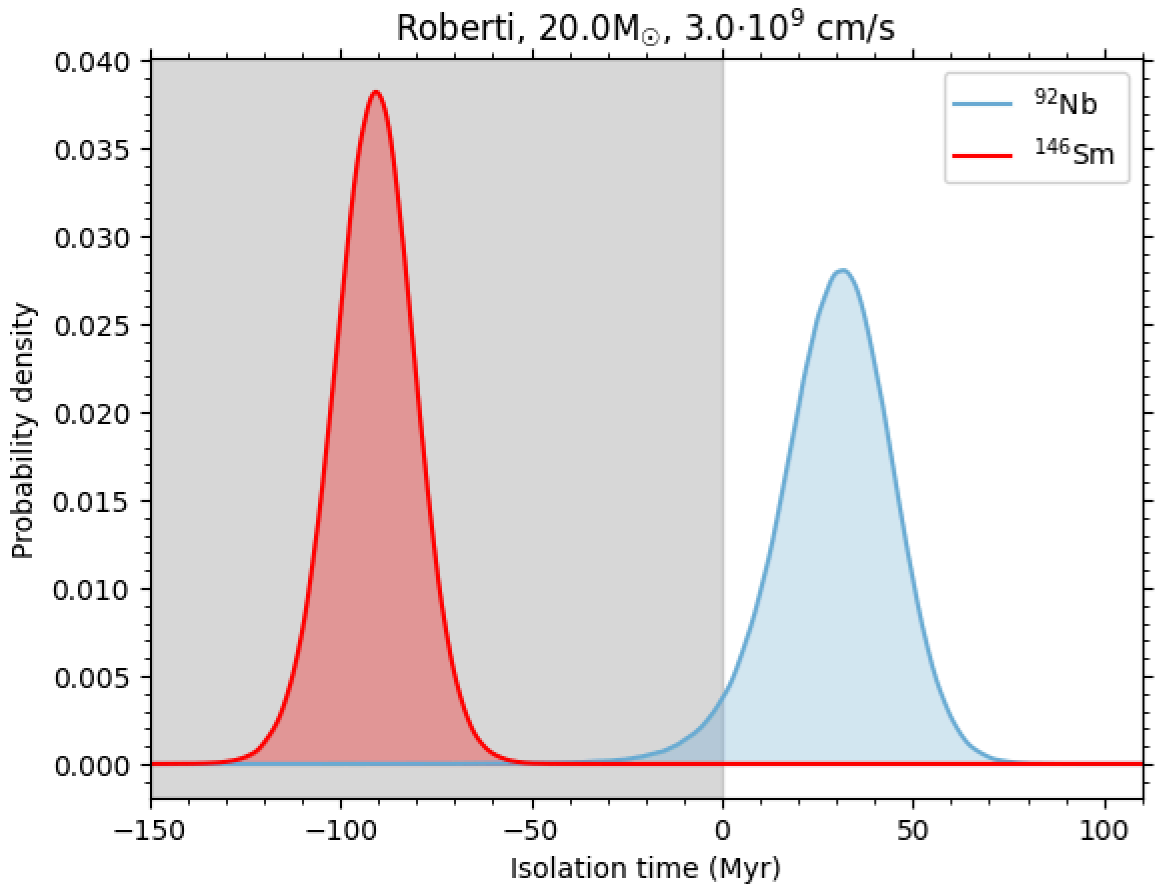
In [30]:
```
#create the ranges for plotting the data (limits selected based on previous
limits = np.array([[-150,50],[-150,50],[-150,110],[-150,110],[-380,110]])
x = np.linspace(limits[:,0], limits[:,1], 1000).T
```

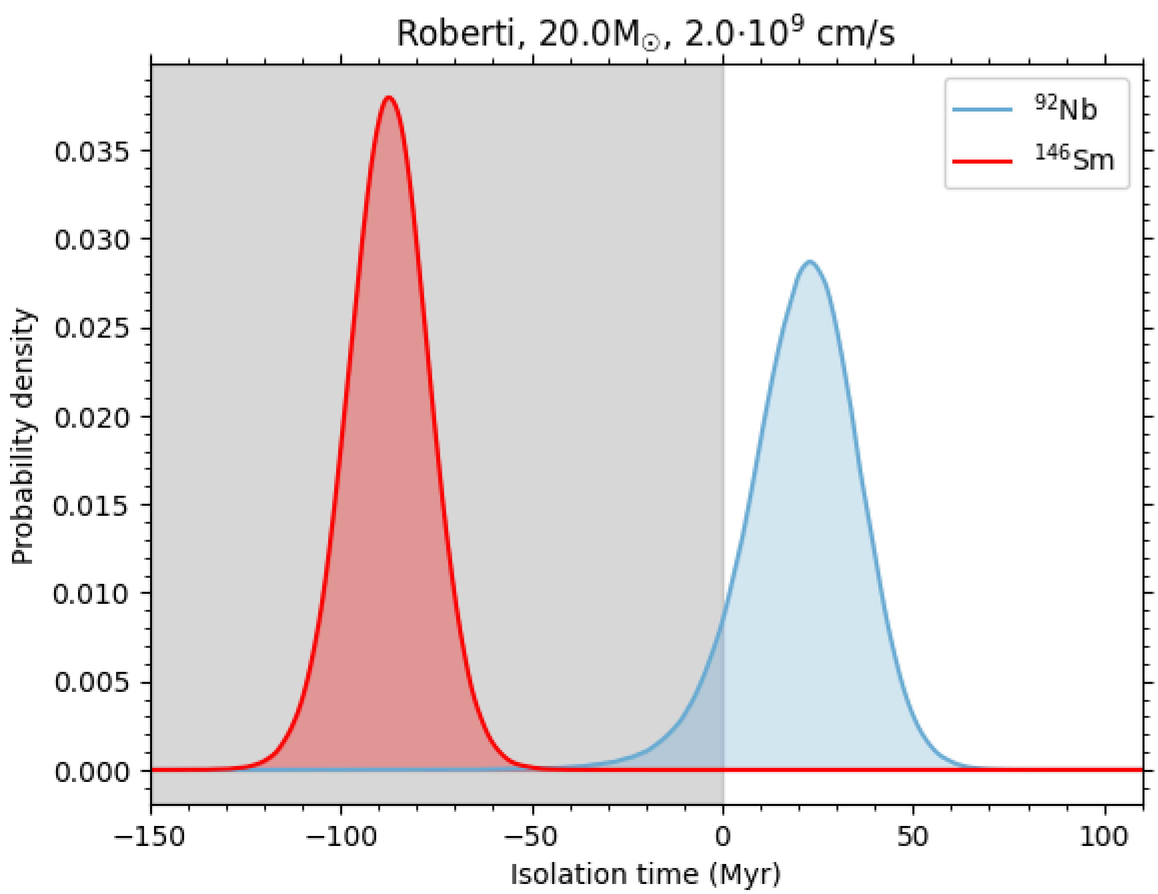
In [31]:
```
#evaluate the distributions at 1000 points
isoTNbDensityBest_CCSn = []
isoTSmDensityBest_CCSn = []

for i in range(np.size(prCCSnNb)):
    isoTNbDensityBest_CCSn.append(np.exp(isoTNbKernCCSn[i].score_samples(x[i
    isoTSmDensityBest_CCSn.append(np.exp(isoTSmKernCCSn[i].score_samples(x[i
```

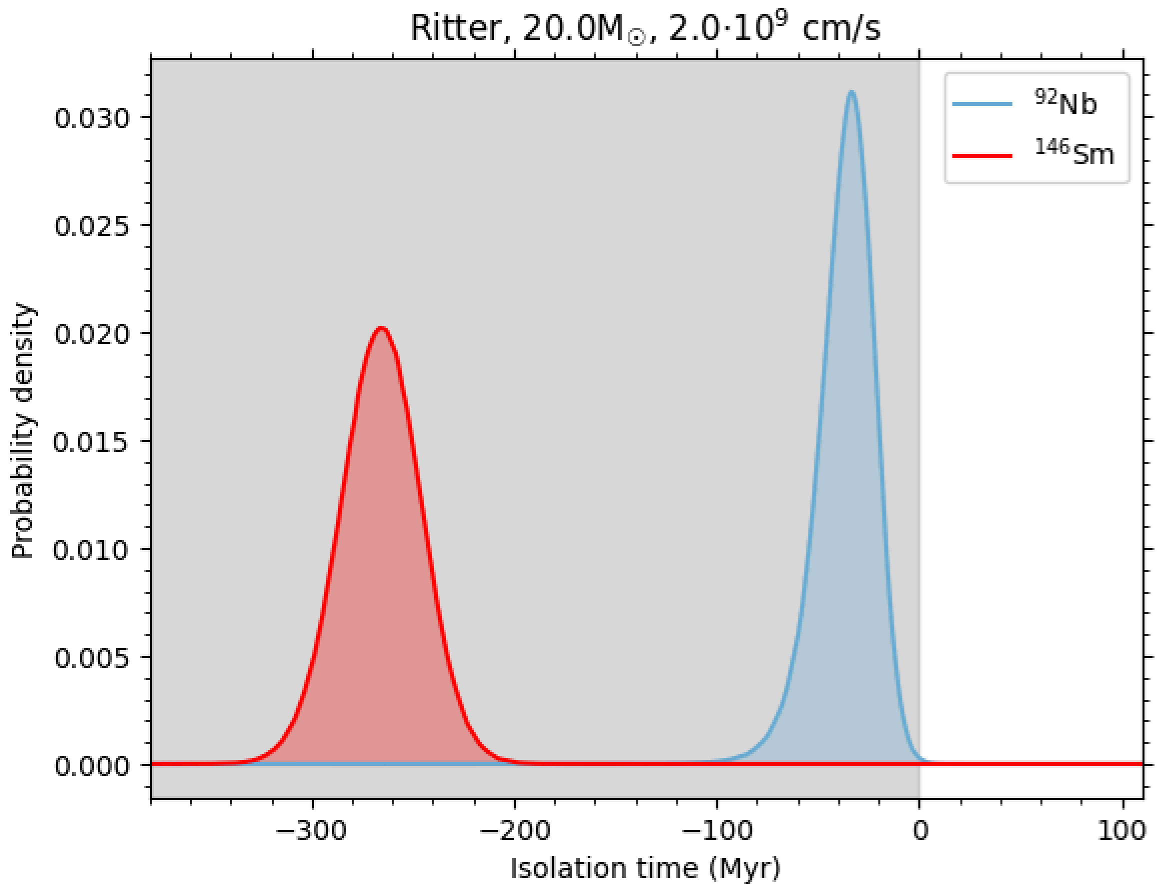
In [32]:
```
blue = plt.get_cmap('Blues')(2/4)

for i in range(np.size(prCCSnNb)):

    plt.plot(x[i], isoTNbDensityBest_CCSn[i], color=blue, label='$^{92}$Nb')
    plt.fill_between(x[i], isoTNbDensityBest_CCSn[i], 0, color=blue,  alpha=

    #for tickmarks
    plt.gca().xaxis.set_minor_locator(AutoMinorLocator())
    plt.gca().yaxis.set_minor_locator(AutoMinorLocator())
    plt.gca().tick_params(axis='x', which='both', top=True)
    plt.gca().tick_params(axis='y', which='both', right=True)

    plt.plot(x[i], isoTSmDensityBest_CCSn[i], color="red", label='$^{146}$Sm
    plt.fill_between(x[i].flatten(), isoTSmDensityBest_CCSn[i], 0, color="re

    plt.xlabel('Isolation time (Myr)')
    plt.ylabel('Probability density')
    plt.legend(loc='upper right')

    #for the gray box
    plt.axvspan(xmin=x[i][0], xmax=0, color='gray', alpha=0.3, zorder=0)
    #plt.text(0.05, 0.95, "Isolation times < 0 \n(grey) are unphysical", tra

    plt.xlim(x[i][0], x[i][-1])

    plt.title(modelSet[i]+", "+str(mtot[i])+"M$_☉$, "+str(shock[i])+"$\\cdot

    plt.savefig("../figures/"+modelSet[i]+"_M"+str(int(mtot[i]))+"_V"+str(sh
    plt.show()
```


Roberti, 20.0M$_\odot$, 5.0·10^9 cm/s
^{92}Nb
^{146}Sm
Probability density
Isolation time (Myr)
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−150
−125
−100
−75
−50
−25
0
25
50


Roberti, 20.0M$_\odot$, 4.0·10^9 cm/s
^{92}Nb
^{146}Sm
Probability density
Isolation time (Myr)
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−150
−125
−100
−75
−50
−25
0
25
50


Roberti, 20.0M☉, 3.0·10^9 cm/s
Probability density
Isolation time (Myr)
92Nb
146Sm
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−150
−100
−50
0
50
100


Roberti, 20.0M☉, 2.0·10^9 cm/s
Probability density
Isolation time (Myr)
92Nb
146Sm
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−150
−100
−50
0
50
100



1σ uncertainty bands for ISS ratios

In [33]:
```
#median, -, +
for i in range(np.size(prCCSnNb)):
    print(modelSet[i]+", "+str(mtot[i])+"M☉, "+str(shock[i])+" · 10⁹ cm/s")
    print("Nb: ",median_and_error(ratIsmNbCCSn[i]))
    print("Sm: ",median_and_error(ratIsmSmCCSn[i]))
    print()
```

```
Roberti, 20.0M☉, 5.0 · 10⁹ cm/s
Nb:  [7.752e-06 -1.925e-06 1.996e-06]
Sm:  [3.391e-03 -4.398e-04 4.694e-04]

Roberti, 20.0M☉, 4.0 · 10⁹ cm/s
Nb:  [7.656e-06 -1.901e-06 1.971e-06]
Sm:  [3.937e-03 -5.106e-04 5.450e-04]

Roberti, 20.0M☉, 3.0 · 10⁹ cm/s
Nb:  [5.871e-05 -1.458e-05 1.512e-05]
Sm:  [3.408e-03 -4.420e-04 4.718e-04]

Roberti, 20.0M☉, 2.0 · 10⁹ cm/s
Nb:  [4.953e-05 -1.230e-05 1.275e-05]
Sm:  [3.534e-03 -4.584e-04 4.893e-04]

Ritter, 20.0M☉, 2.0 · 10⁹ cm/s
Nb:  [1.576e-05 -3.915e-06 4.058e-06]
Sm:  [6.301e-04 -8.173e-05 8.723e-05]
```

1σ uncertainty bands for the Isolation Times

In [34]:
```
#median, -, +
for i in range(np.size(prCCSnNb)):
    print(modelSet[i]+", "+str(mtot[i])+"M☉, "+str(shock[i])+" · 10⁹ cm/s")
    print("Nb: ",median_and_error(isoTNbCCSn[i]))
    print("Sm: ",median_and_error(isoTSmCCSn[i]))
    print()
```

```
Roberti, 20.0M☉, 5.0 · 10⁹ cm/s
Nb:  [-7.055e+01 -1.471e+01 1.194e+01]
Sm:  [-9.151e+01 -1.057e+01 1.032e+01]

Roberti, 20.0M☉, 4.0 · 10⁹ cm/s
Nb:  [-7.117e+01 -1.472e+01 1.195e+01]
Sm:  [-7.626e+01 -1.064e+01 1.078e+01]

Roberti, 20.0M☉, 3.0 · 10⁹ cm/s
Nb:  [3.019e+01 -1.512e+01 1.364e+01]
Sm:  [-9.099e+01 -1.057e+01 1.033e+01]

Roberti, 20.0M☉, 2.0 · 10⁹ cm/s
Nb:  [2.174e+01 -1.496e+01 1.335e+01]
Sm:  [-8.727e+01 -1.057e+01 1.042e+01]

Ritter, 20.0M☉, 2.0 · 10⁹ cm/s
Nb:  [-3.518e+01 -1.449e+01 1.205e+01]
Sm:  [-2.656e+02 -1.985e+01 1.951e+01]
```

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